

Email

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Title	Machine learning-based early diagnosis of disseminated intravascular coagulation by hemostatic biomarkers
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Email	<i>Blood Vessels, Thrombosis & Hemostasis</i> MS# VTH-2025-000444 Title: Machine learning-based early diagnosis of disseminated intravascular coagulation by hemostatic biomarkers Dear Dr. Xu: Expert reviewers in the field have evaluated your Regular Article manuscript, referenced above. Unfortunately, I regret to inform you that our final decision is that we are unable to publish your paper in <i>Blood Vessels, Thrombosis & Hemostasis</i>. While your work is of interest, we did not assign a sufficient priority for publication in <i>Blood Vessels, Thrombosis & Hemostasis</i>. The reviewers brought up various issues and concerns that you can find detailed in their reviews. At <i>Blood Vessels, Thrombosis & Hemostasis</i> we expect that publishable papers should surpass the bar of being either novel or definitive. Our decision is always based on a number of factors some of which are subjective. These do not relate only to the quality and merits of your work but also to our perceived vision of the impact of your work on the field. We greatly appreciate your interest in publishing your work in <i>Blood Vessels, Thrombosis & Hemostasis</i> and regret the negative decision on this manuscript in this instance, but thank you for sending it to us for consideration. We are enclosing the reviewer comments that we hope will be useful to you. Sincerely, Keith R. McCrae Editor-in-Chief, <i>Blood Vessels, Thrombosis & Hemostasis</i> American Society of Hematology Washington, DC

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Reviewer Comments:

Reviewer #1 (Comments to the Author (Required)):

Summary:

Guo et al performed a study to assess biomarkers associated with DIC development in ICU among 700+ patients. overall, TAT / ATIII ratio appears to be associated with other lab findings of overt-DIC and may be helpful to distinguish those who go on to develop DIC vs not. While the overall concept is interesting, there are methodological concerns. It is unclear how the dataset was split into two cohorts (training vs validation). It reports all the models and analyses rather the selective highlights in the tables and figures.

Title:

There is not much machine learning involved, please remove that from the title. Logistic regression alone likely would achieve a similar performance with or without LASSO since there are only 5 variables in the model.

Introduction:

Heavy reliance on AI generated text or revision (e.g. burgeoning field, posited, opportunity for preemptive therapeutic interventions, bridge the critical diagnostic gap, innovative integration, underpinnings, harnessing)

Methods:

Cohort inclusion: need to describe in more detail on how these particular ICU patients were included in the study. Are they consecutive patients admitted to the ICU on day of ICU admission? Are they patients who had a hematology consult? Are there any coagulation issues that led to the consult? Are they simply patients who had leftover samples for hemostatic biomarker testing?

External cohort: need to describe in more detail on this second cohort was created. What is "comprehensive ICU" vs. "emergency ICU" from the same hospital?

Biomarkers: were these measured as part of clinical practice or separately in a research lab setting? Were they measured at a single time point or repeated over time? More details are needed on what assays were used for each assay. For example, there are multiple tests for D-dimer.

Ethics: were the patients consented for the study or did the IRB approve the study without direct patient consent?

Statistical analysis: page 5 lines 131-139, no need to explain what lasso-logit or SHAP is. Describe how they are used for the current analysis. For example, how was the lambda finetuned in the training dataset, was there a priori data split into train / test, what does "five-fold cross-validation to partition the training set for internal validation" mean - is this the outer loop of CV for internal validation ROC, while there is another inner loop CV done to find the optimal lambda? Which model was chosen for the final external validation testing since each CV provides a different model? Was LASSO only used for feature selection which then led to model refitting in a regular Logit for AUC estimation?

Results:

Overall, would suggest finding one key outcome rather than multiple subgroup analysis. For example, the prediction of DIC development over 7 days (yes vs no) is sufficient (no need to split further into 1-3 vs 4-7).

Table 2-3. Suggest combining into 1 table; for each half of the table, simplify to 4 columns: non-DIC, pre-DIC, over-DIC, and one single p-value column for multi-group comparison if desired

Table 4. Move to supplemental

Table 5 (and corresponding portions in Results): please highlight the key findings (ideally the best model, not all models) and remove all the extra comparisons because the presentation is very confusing for readers. On day 0, the comparison was for DIC vs none. On day 1-7, the comparison was for those who eventually developed DIC vs none. Can remove day 1-3 vs 4-7 sub-comparison for simplicity.

Table 6-8 (and corresponding portions in Results): likely unnecessary for most readers

Table 9. keep first two columns and remove second two columns. Remove recall (recall is the same as sensitivity), change precision to positive predictive value, add negative predict value as well, remove F1-

score.

Table 9. did the model require all the unusual biomarkers or could it have performed similarly well with basic ones like platelet count, ATIII, PT, fibrinogen, D-dimer, and FDP? It would be interesting to see if the "novel" biomarkers offered any additional predictive value.

Table 9 (and corresponding portions in Results): what are the beta coefficients (OR) of the covariates in the actual final model? The point of LASSO is penalty shrinkage to remove non-significant predictors. Were all predictors kept in the final model?

Figure 2. remove

Figure 3. move to supplemental

Figures 4a-d. remove; don't need SHAP values to interpret a standard logistic regression model. please provide OR instead per above (penalized OR is fine too).

Discussion:

Can likely reduce word count by 50%; highlight 2-3 most important novel concepts and comparison to previous studies

Recommend removal of AI generated language (e.g. crucial, innovative, invaluable, foresight, empowering, revolutionize, important discoveries, inherently, robust, signifies, the entire conclusion paragraph)

Reviewer #2 (Comments to the Author (Required)):

The predictive model presented relies almost exclusively on specialized biomarkers such as thrombin-antithrombin complex, thrombomodulin, tissue plasminogen activator inhibitor complex, and plasmin- α 2-plasmin inhibitor complex. While mechanistically interesting, these assays are not part of routine patient care. Their absence from standard practice is not a matter of oversight but reflects fundamental issues of accessibility, cost, and turnaround time. At best, they can be justified as exploratory endpoints in small mechanistic studies. To propose them as the foundation of a predictive model in critically ill patients is unrealistic, as the vast majority of ICUs-even at major tertiary centers-lack the ability to measure them in real time. This reliance on esoteric testing severely undermines the translational value of the work.

The claim that these biomarkers are "routinely measured" is inaccurate and not used to guide patient management. They may be ordered under research protocols, but their clinical use is virtually nonexistent. Even in those limited research contexts, their acquisition requires considerable resources. The assays themselves are costly, the platforms are not broadly distributed, and the turnaround times are incompatible with the rapid decision-making required in critically ill patients. These realities cannot be dismissed, and they fundamentally weaken the assertion that the findings are generalizable.

Beyond issues of laboratory feasibility, the authors appear to treat disseminated intravascular coagulation as a primary condition suitable for prediction in isolation. In reality, DIC is a secondary syndrome, always driven by an underlying disease process. In the ICU, sepsis is the most common and clinically relevant precipitant, although trauma, malignancy, and obstetric catastrophes are also recognized causes. The onset and progression of DIC are dictated by the dynamics of the underlying condition and the host response, both of which are highly variable. By abstracting DIC from its causative illness, the model ignores the central determinant of patient trajectory. This conceptual error further undermines both the predictive accuracy and clinical relevance of the work.

The suggestion that early changes in these biomarkers could guide intervention also lacks credibility. While it is theoretically appealing to propose that subtle laboratory changes may herald the onset of clinically relevant coagulopathy, there is no evidence that detecting these changes earlier leads to better outcomes. Indeed, no therapeutic strategy has been proven to alter the course of DIC once it is recognized, let alone at the stage of subclinical biomarker abnormalities. Without a defined intervention that demonstrably improves outcomes, the utility of earlier detection remains speculative. To promote such a model risks implying a clinical actionability that does not exist.

Equally concerning is the treatment of statistical association as if it implies causation. For example, the thrombin-antithrombin/antithrombin III ratio is proposed as a threshold for possible intervention. This rests entirely on observed correlations, not mechanistic validation or interventional evidence. Previous studies

	have explored antithrombin levels extensively, with consistent demonstration of association but no proof that therapeutic manipulation improves outcomes. Without causal evidence, such thresholds are at best descriptive.
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